Solution

Explore Introduction to LTI Systems

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AllSignalProcessing.com



Explore Intro to LTI Systems

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Set Up Workspace

```
clear
close all
set(0,'defaultaxesfontsize',18);
set(0,'defaulttextfontsize',18);


[CfA,fs] = audioread('CarefreeA.wav');

soundsc(CfA,fs)
pause   % Hit any key in the workspace to continue executing the code
```

Part 1

System 1 passes low frequencies and system 2 passes high frequencies.

```
A1 = [1 -2.6862 2.4197 -.7302];
B1 = 0.001*[.4165 1.2496 1.2496 .4165];

A2 = [1.0000   -1.7600    1.1829   -0.2781];
B2 = [0.5276   -1.5829    1.5829   -0.5276];

CfA1 = filter(B1,A1,CfA);
soundsc(CfA1,fs)
pause

CfA2 = filter(B2,A2,CfA);
soundsc(CfA2,fs)
```

Part 2

Part a) System 1 has a smoother impulse response, consistent with its selection of lower frequencies.

Part b) System 1 passes very low frequencies (below about 0.1 rads) and attenuates high frequencies (above about 0.5 rads). System 2 attenuates low frequencies (say below about 0.2 rads) and passes high frequencies (say above about 1 rad). This

agrees with the low/high freq behavior observed in part 1.

```
% Part a

imp = [1 zeros(1,119)];

h1 = filter(B1,A1,imp);

figure
stem([0:119],h1)
xlim([0,119])
xlabel('n')
ylabel('Amplitude')
title('System 1 Impulse Response')

h2 = filter(B2,A2,imp);

figure
stem([0:119],h2)
xlim([0,119])
xlabel('n')
ylabel('Amplitude')
title('System 2 Impulse Response')

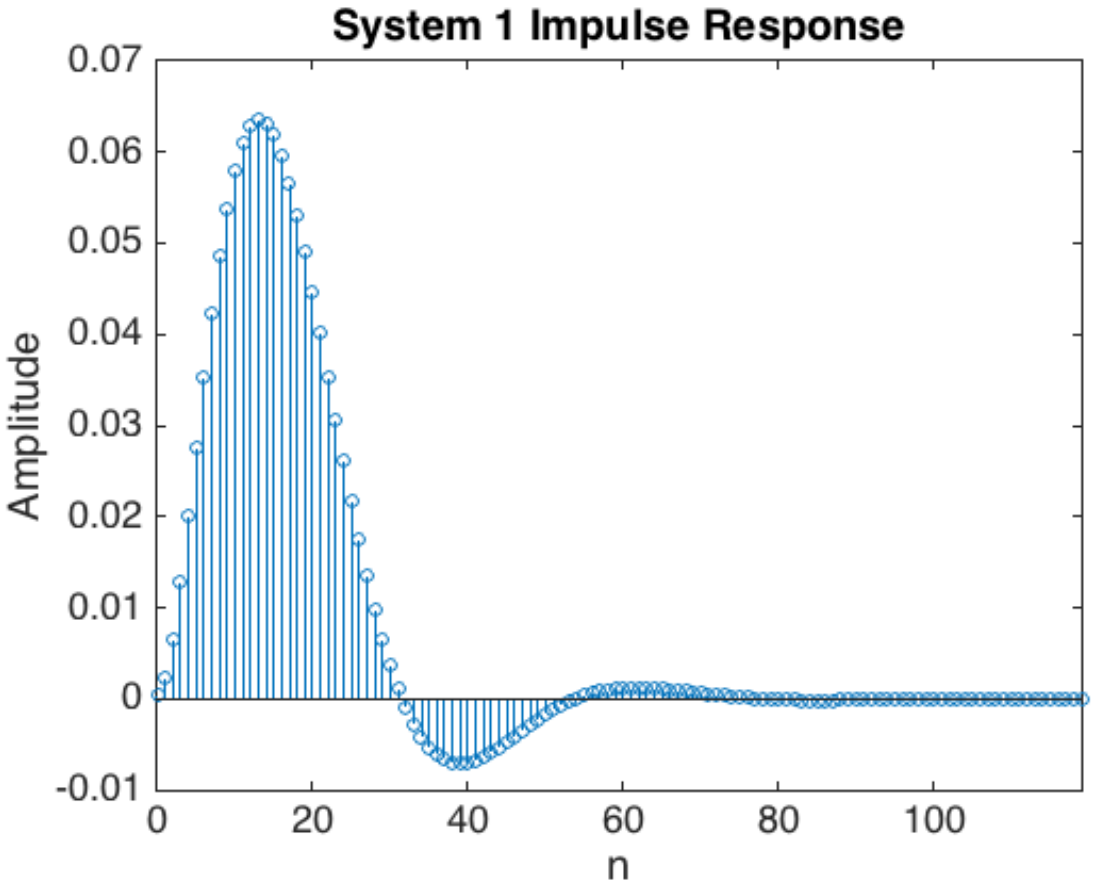
% Part b

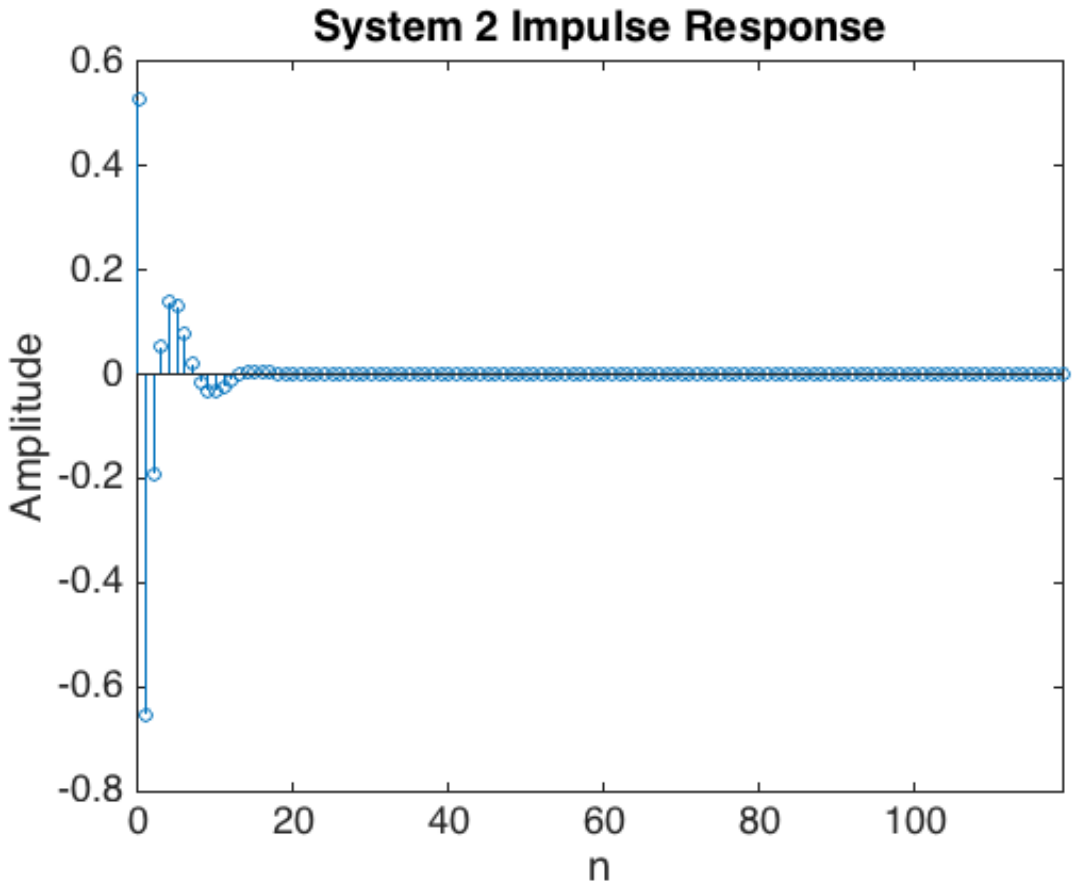
[H1,om] = freqz(B1,A1);
figure
plot(om,abs(H1))
xlim([0 pi])
xlabel('Frequency (rads)')
ylabel('Gain')
title('System 1 Magnitude Response')

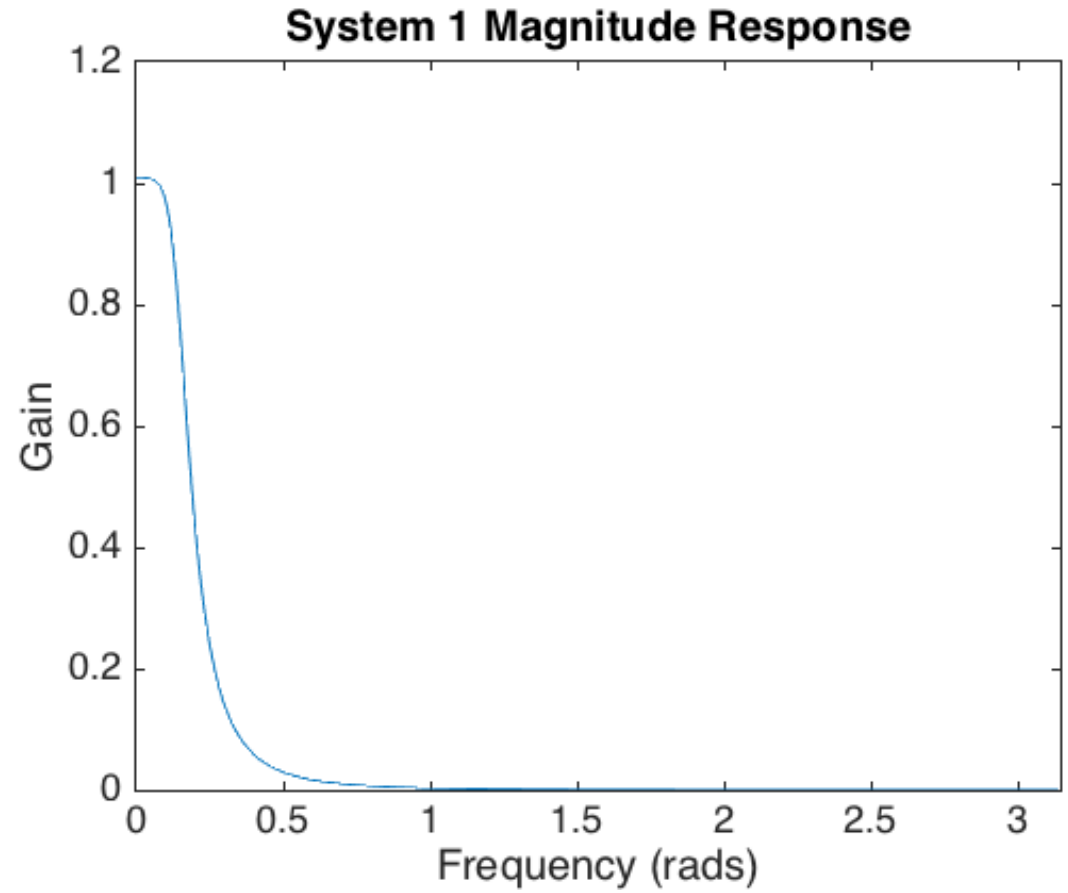
figure
plot(om,phase(H1))
xlim([0 pi])
xlabel('Frequency (rads)')
ylabel('Phase (rads)')
title('System 1 Phase Response')

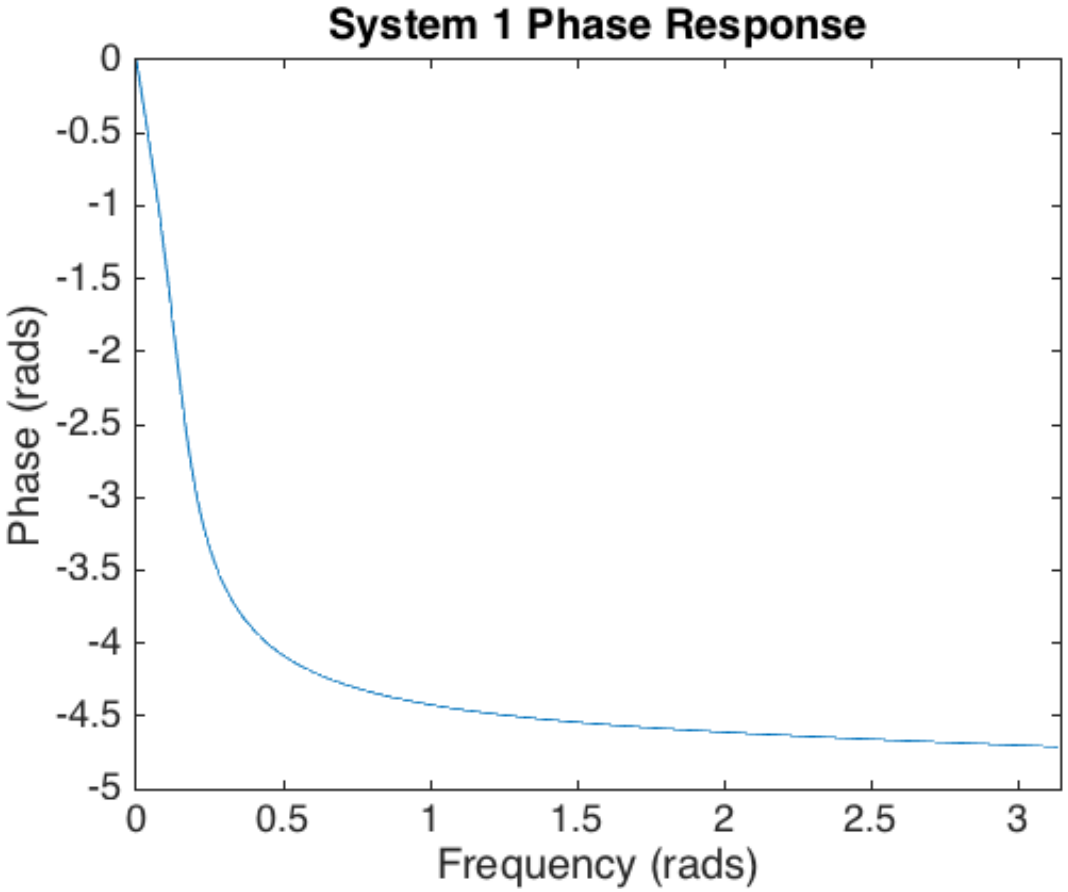

[H2,om] = freqz(B2,A2);
figure
plot(om,abs(H2))
xlim([0 pi])
xlabel('Frequency (rads)')
ylabel('Gain')
title('System 2 Magnitude Response')

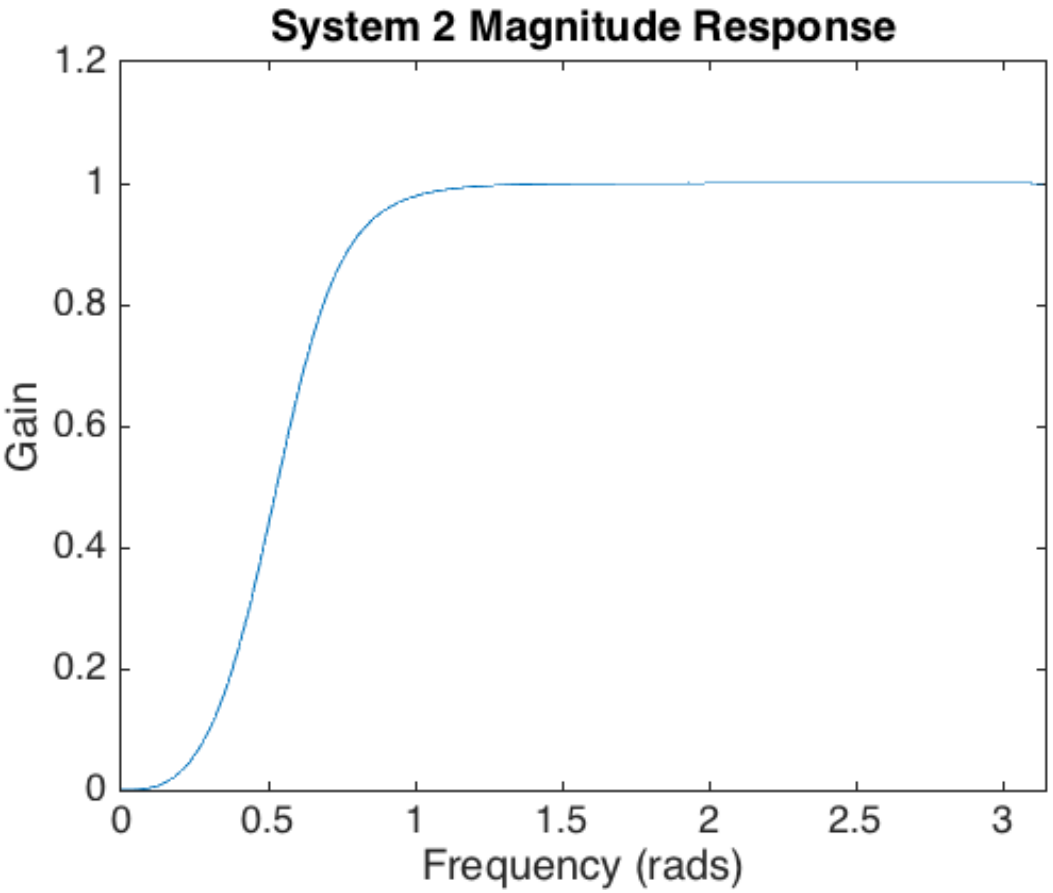
figure
plot(om,phase(H2))
xlim([0 pi])
xlabel('Frequency (rads)')
ylabel('Phase (rads)')
title('System 2 Phase Response')
```

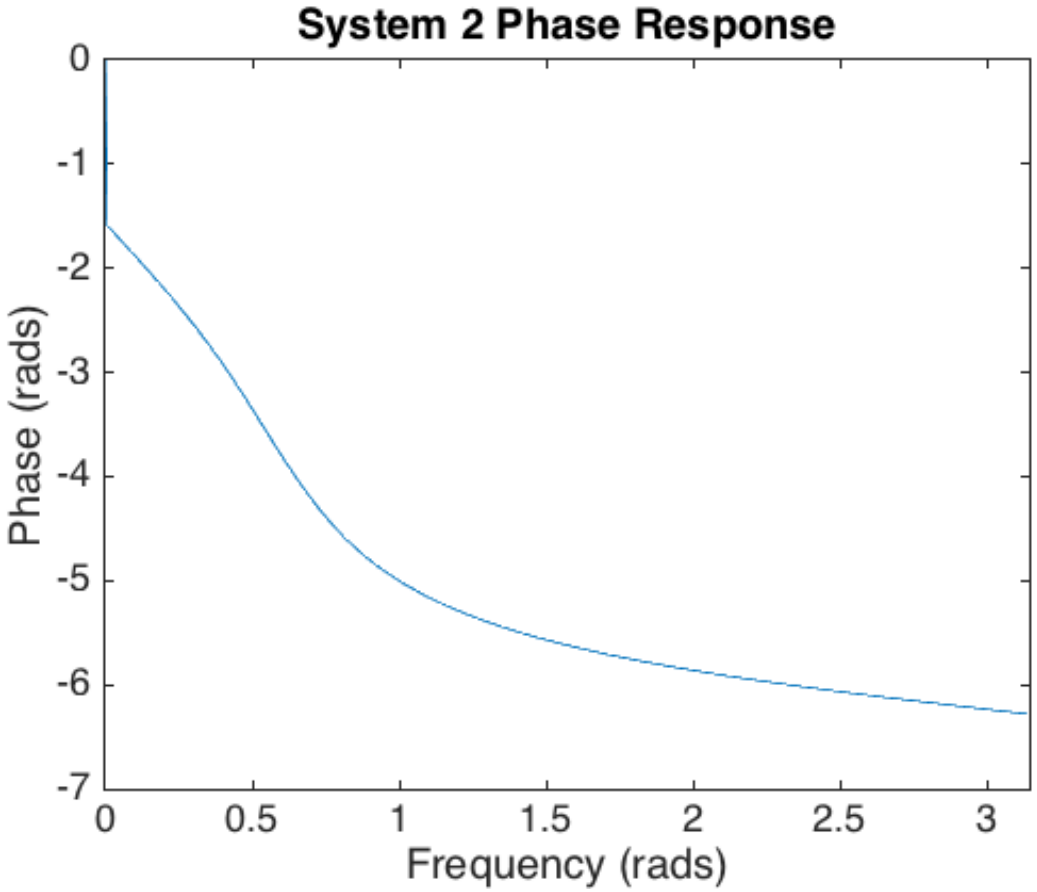

System 1 Impulse Response
Amplitude
n
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
-0.01
0
20
40
60
80
100


System 2 Impulse Response
Amplitude
n
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
20
40
60
80
100


System 1 Magnitude Response
Gain
Frequency (rads)
0
0.2
0.4
0.6
0.8
1
1.2
0.5
1.5
2
2.5
3


System 1 Phase Response
Phase (rads)
Frequency (rads)
0
-0.5
-1
-1.5
-2
-2.5
-3
-3.5
-4
-4.5
-5
0.5
1
1.5
2
2.5
3


System 2 Magnitude Response
Gain
Frequency (rads)
0
0.2
0.4
0.6
0.8
1
1.2
0.5
1.5
2
2.5
3


System 2 Phase Response
Phase (rads)
Frequency (rads)
0
-1
-2
-3
-4
-5
-6
-7
0.5
1
1.5
2
2.5
3

Part 3

Once in a steady state condition, system 1 passes the low frequency cosine and attenuates the high frequency cosine, while system 2 attenuates the low frequency cosine and passes the high frequency cosine. Note that the gain of system 1 at pi/50 rads is ~ 1, while at pi/3 rads it is ~ 0. System 2 has a gain of ~ 0 at pi/50 rads and ~ 1 at pi/3 rads.

The impulse response and convolution produce identical outputs as the difference equation.

```
% First part

cosLF = cos([0:249]*pi/50);
cosHF = cos([0:249]*pi/3);

ydiff1LF = filter(B1,A1,cosLF);
ydiff1HF = filter(B1,A1,cosHF);

figure(7)
subplot(211)
stem([0:249],ydiff1LF)
ylabel('Amplitude')
xlabel('n')
title('Low Freq Cos: System 1')

subplot(212)
stem([0:249],ydiff1HF)
ylabel('Amplitude')
xlabel('n')
title('High Freq Cos: System 1')

ydiff2LF = filter(B2,A2,cosLF);
ydiff2HF = filter(B2,A2,cosHF);

figure(8)
subplot(211)
stem([0:249],ydiff2LF)
ylabel('Amplitude')
xlabel('n')
title('Low Freq Cos: System 2')

subplot(212)
stem([0:249],ydiff2HF)
ylabel('Amplitude')
xlabel('n')
title('High Freq Cos: System 2')

% Second part

yh1LF = conv(h1,cosLF);
yh1HF = conv(h1,cosHF);

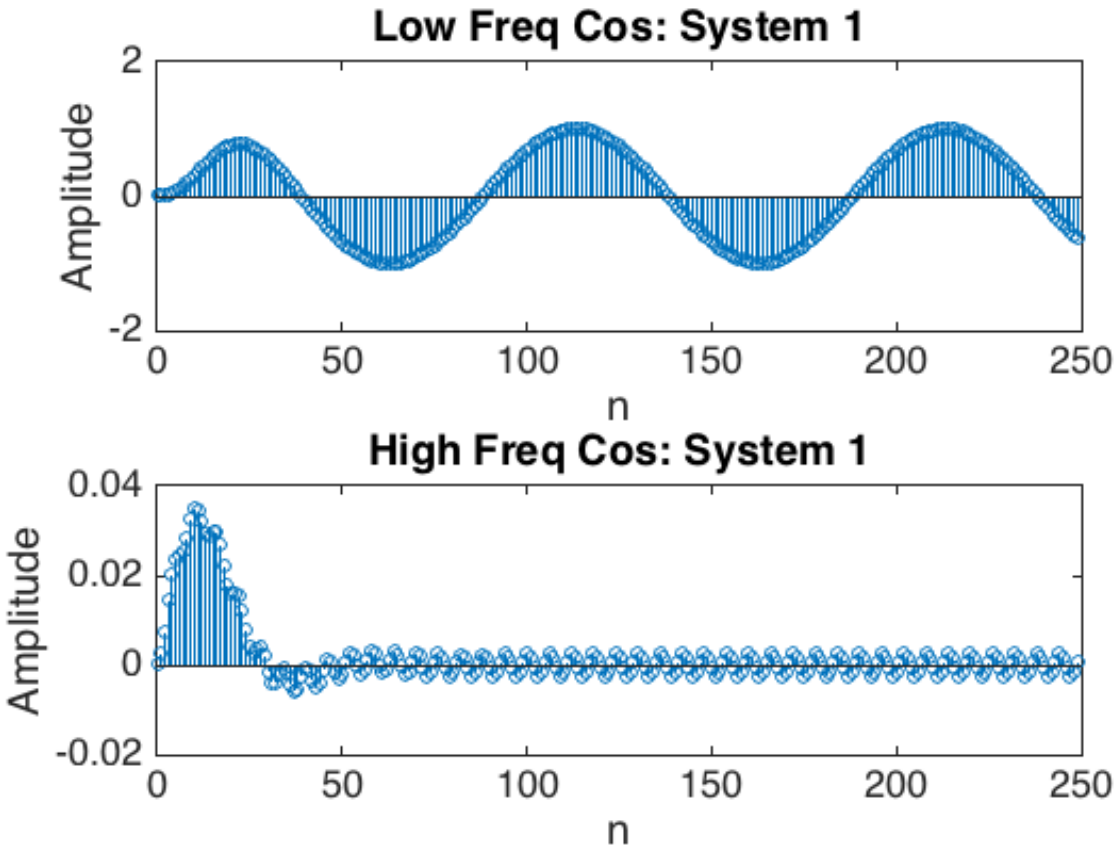
figure(9)
subplot(211)
stem([0:249],ydiff1LF)
ylabel('Amplitude')
xlabel('n')
title('Low Freq Cos: Syst 1 Diff Eqn')

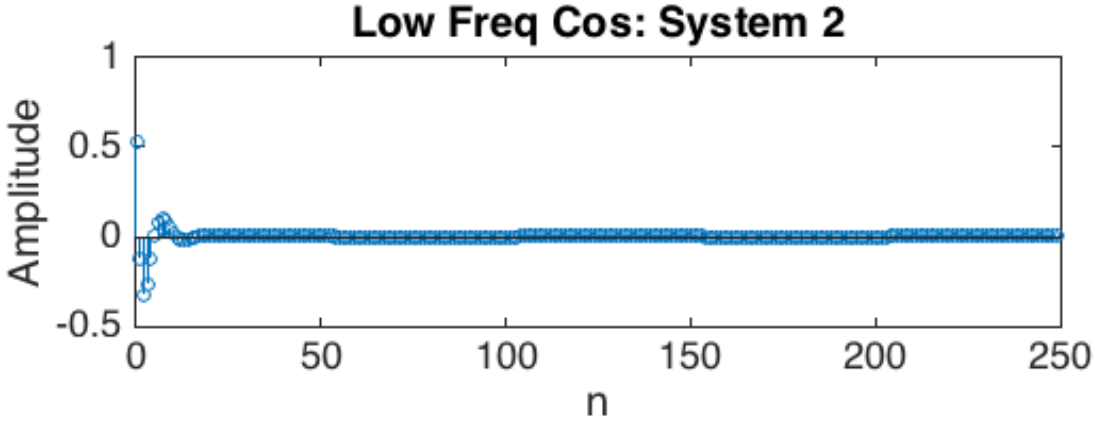
subplot(212)
```

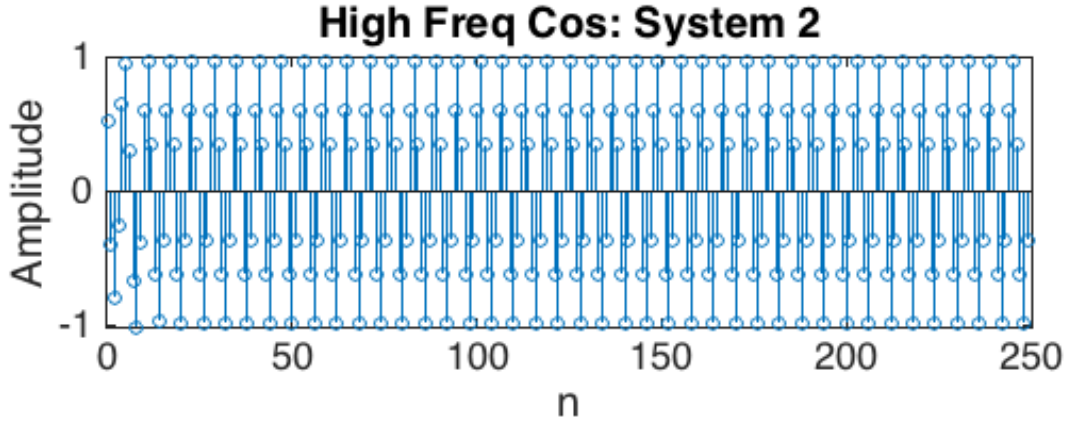
```
stem([0:249],yh1LF(1:250))
ylabel('Amplitude')
xlabel('n')
title('Low Freq Cos: Syst 1 Imp Resp')

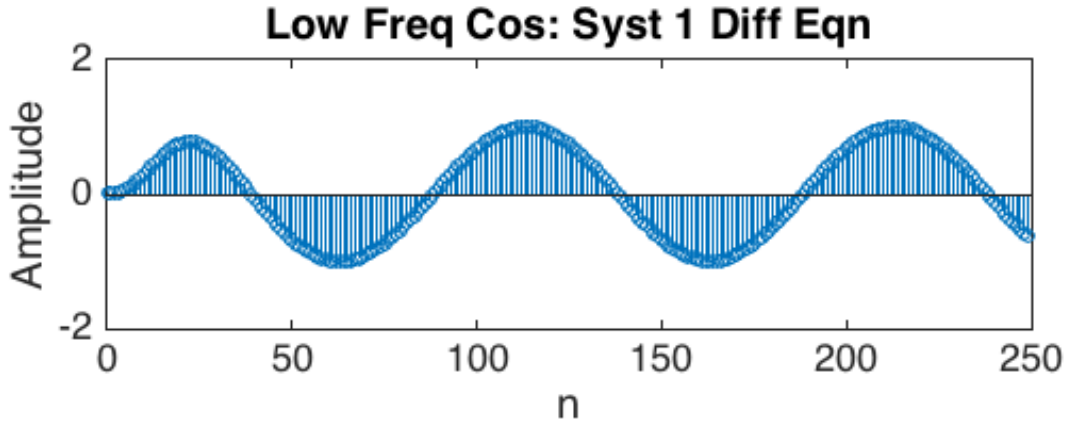
figure(10)
subplot(211)
stem([0:249],ydiff1HF)
ylabel('Amplitude')
xlabel('n')
title('High Freq Cos: Syst 1 Diff Eqn')

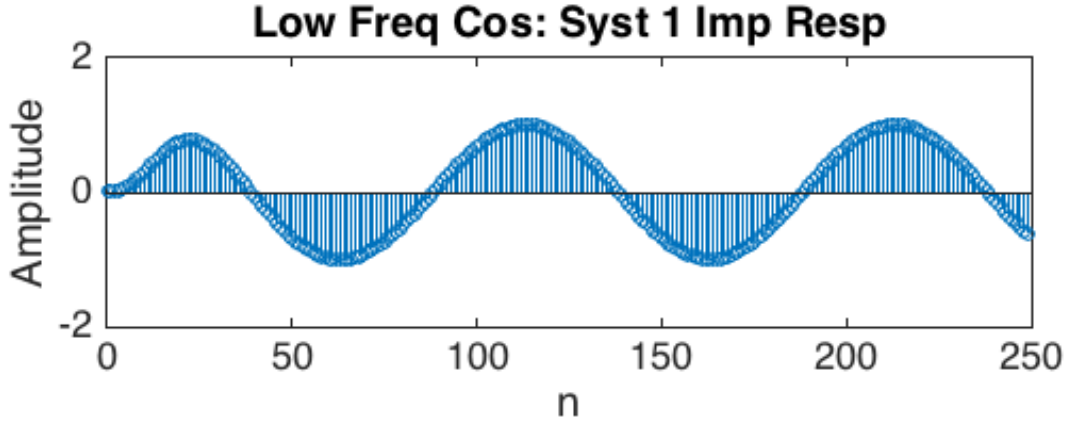
subplot(212)
stem([0:249],yh1HF(1:250))
ylabel('Amplitude')
xlabel('n')
title('High Freq Cos: Syst 1 Imp Resp')
```

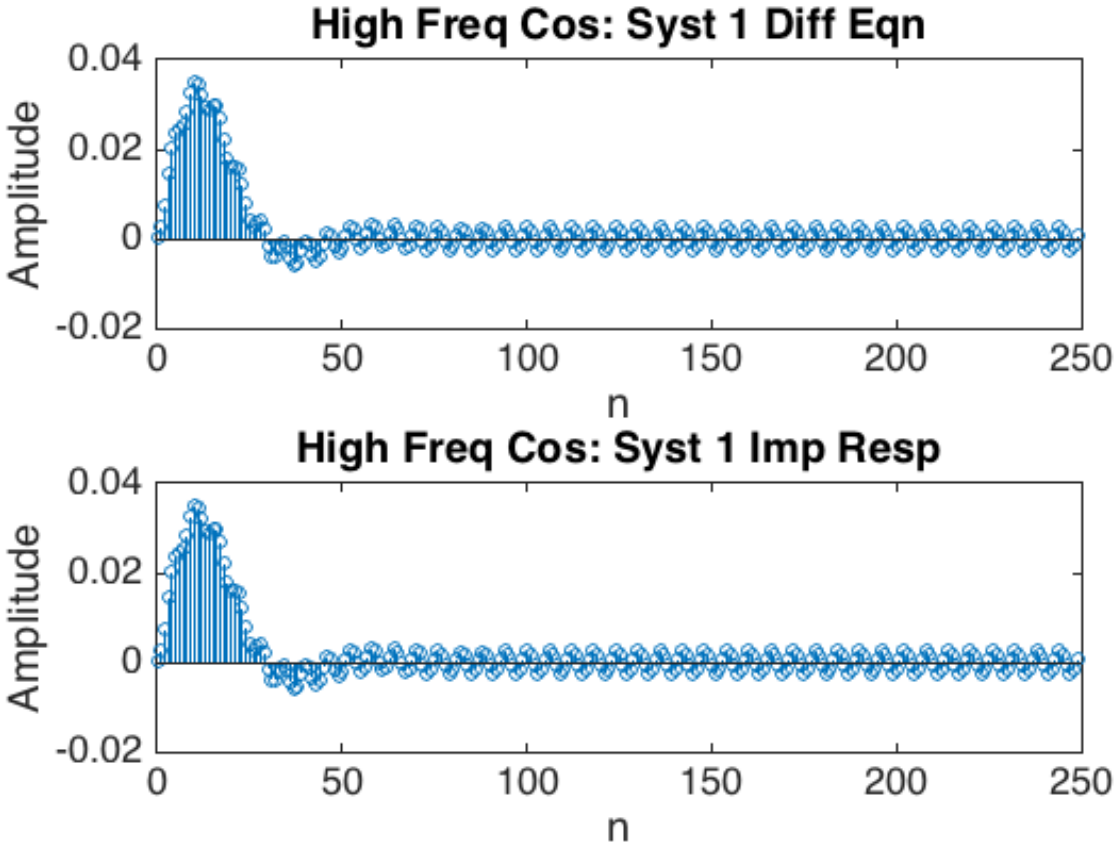



Low Freq Cos: System 2
Amplitude
n
1
0.5
0
-0.5
0
50
100
150
200
250


High Freq Cos: System 2
Amplitude
n
1
0
-1
0
50
100
150
200
250


Low Freq Cos: Syst 1 Diff Eqn
Amplitude
n
2
0
-2
0
50
100
150
200
250


Low Freq Cos: Syst 1 Imp Resp
Amplitude
n
2
0
-2
0
50
100
150
200
250



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